

IIT JODHPUR

Feb'26 Minor Examination: EEL71020 Hardware Design for AI

Guidelines (Total time: 120 minutes, Maximum Marks: 24):

- Please read the question paper very carefully. **NO clarification is required in any question.** In case of any doubt, assume whatever you wish to and state that in your answer.
 - Usage of LLM/AI tool is NOT allowed. BOOKS/NOTES etc. can be referred to. For Offline students, access to LAPTOP/MOBILE is NOT allowed.
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1. Consider that you are tasked with designing a Multiply-Accumulate (MAC) unit for a neural network accelerator targeting edge inference tasks in a deep-tech startup *LekhakAI* primarily involving in developing lightweight text-based applications. The neural network layer requires: 256 input neurons, 128 output neurons, 8-bit signed weights and 8-bit signed activations and 32-bit accumulation. Each MAC unit occupies 0.030 mm^2 area, has a latency of 2 clock cycles per MAC operation and is fully pipelined and therefore it can accept new inputs every cycle. You are given an area budget of 6 mm^2 area for the MAC units only and it is provided that the system clock frequency is 500 MHz. [Hint: We use 32-bit accumulation as the hardware design must be safe for all input sizes (worst-case etc.)].
 - Calculate total number of MAC operations required to compute one forward pass of this layer? [1]
 - Assuming optimal scheduling and full utilization of MAC units, calculate the number of clock cycles required to compute the layer. [1.5]
 - If the system must support 10,000 inferences per second, determine whether the current architecture meets the throughput requirement? How? [2.5]
 - Briefly explain how reducing precision from 8-bit to 4-bit would impact the area? [1]
2. You are an intern in *FraudRoko* startup that deals with secure financial transaction system design. As a part of the custom silicon design team there, you are asked to implement a Random Forest (RF) classifier for tabular data analytics in the company's flagship financial fraud detection system. You are provided with the following specification of random forest (RF)-based classification model and the specifications of the hardware platform (as shown in the right):

Number of trees = 64 Maximum depth per tree = 6 Each internal node stores: • Feature index $\rightarrow$ 6 bits (supports up to 64 features) • Threshold value $\rightarrow$ 12 bits (fixed-point representation) Each leaf node stores: Class label $\rightarrow$ 3 bits (supports up to 8 classes)	Take these assumptions: • A full binary tree of depth 6 is used. • All parameters are stored in on-chip SRAM. • The memory is byte-addressable. • The system clock is 250 MHz. • One node is accessed per cycle. • The memory block size must be a power of 2. • Only one memory read port is available.
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For the above RF implementation, design an address decoder that: selects one of 64 trees and also selects a node inside that tree. Also, determine the number of address bits required to: i) Select tree ID ii) Select node index within a tree iii) Address the entire memory [2 + 3]

3. Suppose that you incubate a startup called as *SmartDrishtiTech* at IIT Jodhpur. As a team leader in *SmartDrishtiTech*, you are tasked with designing a hardware accelerator for implementing a small Convolutional Neural Network (CNN) on an edge device used for real-time object detection in a battery-powered smart camera. Your edge device has the following constraints:

- On-chip SRAM: 512 KB
- No external DRAM access allowed (this is to save power)
- Maximum power budget: 1.5 W
- Target latency per image: < 30 ms
- 8-bit quantized weights and activations

After the training process carried out with a specially curated city-specific dataset, the model consists of a structure follows: CNN with 3 convolution layers (3×3 kernels), 1 fully connected layer, ReLU activations and Max pooling after the first two convolution layers [Hint: Max pooling means finding the maximum element out of the total elements in a 2x2 matrix, 3x3 matrix and so on].

- How you would organize memory for weights and feature maps within 512 KB SRAM? [2]
 - Mention few techniques to reduce memory access energy in your edge device design? [2]
 - How you can achieve parallelism in accelerator design? Explain your design in a concise manner. [2]
 - How quantization (8-bit) impacts area, power, and throughput of your design? [1]
4. You are a design engineer in a semiconductor hardware startup that sells custom silicon products to different clients. One of such clients of this startup is an edtech company *PathsalAI* that plans to deploy a lightweight text-based classification model based on Principal Component Analysis (PCA) algorithm. You are tasked with designing an on-chip hardware accelerator to compute the correlation matrix required for PCA implementation. The targeted system specifications are as follows:

Number of input features (dimensions) = 16 Number of samples in dataset (N) = 512 Each feature value is represented in 12-bit signed fixed-point format Correlation values must be computed in 32-bit fixed-point format	System clock frequency = 400 MHz Available on-chip SRAM = 64 KB Only one read and one write port per SRAM bank Maximum allowed computation latency = 50 μs
The correlation matrix R is defined as: $R(i, j) = \frac{1}{N} \sum_{k=1}^{512} X_i(k) \cdot X_j(k)$	

- Given that you are allowed to instantiate up to 32 MAC units, propose a datapath architecture for following different styles of implementation: a) Fully parallel, b) Partially parallel [4]
- Compute the total number of cycles required if: (a) 32 MAC units operate in parallel (b) One multiplication per MAC happens per cycle [2]